

LIBRARY MANAGEMENT SYSTEM

Description

A Java console application that manages library operations using JDBC and MySQL.

Features

- Add Book
- View Books
- Update Book
- Delete Book
- Add User
- Borrow Book
- Return Book

Technologies Used

- Java
- JDBC
- MySQL

Concepts Used

- JDBC
- CRUD Operations
- Prepared Statement
- SQL
- Exception Handling

Database

MySQL

Database Name:

...

library_db

...

How to Run

1. Create the MySQL database.
2. Configure `DBConnection.java`.
3. Run `Main.java`.

Sample Output

===== Library Management System =====

1. Add Book
2. View Books
3. Update Book
4. Delete Book
5. Add User
6. Borrow Book
7. Return Book
8. Exit

Enter your choice: 1

Enter Book ID: 101

Enter Book Name: Java Programming

Enter Author Name: Herbert Schildt

Enter Price: 450

Book added successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Update Book
4. Delete Book
5. Add User
6. Borrow Book
7. Return Book
8. Exit

Enter your choice: 1

Enter Book ID: 102

Enter Book Name: Database Management System

Enter Author Name: Korth

Enter Price: 550

Book added successfully!

===== Library Management System =====

1. Add Book

2. View Books
3. Update Book
4. Delete Book
5. Add User
6. Borrow Book
7. Return Book
8. Exit

Enter your choice: 2

===== Available Books =====

Book ID	Book Name	Author	Price
101	Java Programming	Herbert Schildt	450.0
102	Database Management System	Korth	550.0

===== Library Management System =====

1. Add Book
2. View Books
3. Update Book
4. Delete Book
5. Add User
6. Borrow Book
7. Return Book
8. Exit

Enter your choice: 5

Enter User ID: 201

Enter User Name: Ravi Kumar

Enter Email: ravi@gmail.com

User added successfully!

===== Library Management System =====

1. Add Book
2. View Books
3. Update Book

4. Delete Book

5. Add User

6. Borrow Book

7. Return Book

8. Exit

Enter your choice: 6

Enter User ID: 201

Enter Book ID: 101

Book borrowed successfully!

===== Library Management System =====

1. Add Book

2. View Books

3. Update Book

4. Delete Book

5. Add User

6. Borrow Book

7. Return Book

8. Exit

Enter your choice: 7

Enter User ID: 201

Enter Book ID: 101

Book returned successfully!

===== Library Management System =====

1. Add Book

2. View Books

3. Update Book

4. Delete Book

5. Add User

6. Borrow Book

7. Return Book

8. Exit

Enter your choice: 3

Enter Book ID to update: 102

Enter New Book Price: 600

Book updated successfully!

===== Library Management System =====

1. Add Book

2. View Books

3. Update Book

4. Delete Book

5. Add User

6. Borrow Book

7. Return Book

8. Exit

Enter your choice: 4

Enter Book ID to delete: 102

Book deleted successfully!

===== Library Management System =====

Enter your choice: 8

Thank you for using Library Management System.

Application Closed Successfully...

Author

Gopavaram Suchithra.